Arista Access Node WFD-ACC-44 Optical Transport Failure SOP (ALARM_8797)

Node: WFD-ACC-44 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_8797

Description

This Standard Operating Procedure (SOP) is designed to address major optical transport failures on the Arista access node WFD-ACC-44 at the Whitefield POP. The failure domain in question is precipitated by SFP/XFP degradation, fiber cuts, or Rx/Tx power loss, which can compromise the integrity of the transport layer and disrupt data transmission. This phenomenon occurs due to the node's role as an access point, where it facilitates communication between the network core and the customer premises equipment (CPE). The Arista EOS (Extensible Operating System) relies on the SFP/XFP to establish and maintain connections, and any degradation in the optical signal can trigger an alarm. The ALARM_8797 code specifically indicates a major failure in the optical transport domain, necessitating prompt action to prevent network outages and data losses.

Pre-requisite Checks

- check: Verify the site location and node ID to ensure it matches the failure context., result: Expected: WFD-ACC-44 at Whitefield POP.

- check: Confirm the CLI dialect is set to Arista EOS., result: Expected: Arista EOS CLI dialect confirmed.

- check: Ensure redundant link status is checked before proceeding., result: Expected: Redundant link status confirmed operational.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces status, Expected Output: Interface Status: up/down - last flapped 00:00:00 ago, Description: This command isolates the interface status, which can indicate if the issue is related to the SFP/XFP or fiber cuts.

Command: show inventory, Expected Output: Inventory: SFP/XFP - serial number - revision - vendor - model, Description: This command provides detailed information about the SFP/XFP, including the serial number, revision, vendor, and model, which can help identify potential issues with the device.
```

```
Command: show log | grep ALARM_8797, Expected Output: ALARM_8797: Optical transport failure detected, Description: This command searches the system logs for the ALARM_8797 code and provides information about the optical transport failure.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP, Command: replace SFP/XFP , Description: Replace the faulty SFP/XFP with a new one and verify the interface status.

Step 2: Step: Verify fiber connectivity, Command: show interfaces status, Description: Verify the fiber connectivity and ensure it is properly connected to the node.

Step 3: Step: Adjust Rx/Tx power settings, Command: set interfaces rx-power-adjust , Description: Adjust the Rx/Tx power settings to ensure optimal signal strength and prevent further issues.

Post-Check Verification

```
Command: show interfaces status, Expected Output: Interface Status: up - last flapped 00:00:00 ago, Description: Verify that the interface status is up and the issue has been resolved.

Command: show log | grep ALARM_8797, Expected Output: No output, Description: Verify that the ALARM_8797 code is no longer present in the system logs.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: replace SFP/XFP , Description: Restore the original SFP/XFP and verify the interface status.

- Step: Revert Rx/Tx power settings, Command: set interfaces rx-power-adjust , Description: Revert the Rx/Tx power settings to their original values.